

# A Paley–II analogue of the Scarpis–Đoković construction for prime powers $q \equiv 1 \pmod{4}$

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June 16, 2026

## Abstract

A Hadamard matrix of order  $m$  is a  $\{1, -1\}$ -matrix  $H$  with  $HH^T = mI_m$ . Scarpis (1898) showed that for a prime  $p \equiv 3 \pmod{4}$  a Hadamard matrix of order  $p + 1$  can be lifted to one of order  $p(p + 1)$ , and Đoković extended the lift to all prime powers  $q \equiv 3 \pmod{4}$ , leaving open the analogous procedure for  $q \equiv 1 \pmod{4}$ . We give an explicit, uniform construction of a Hadamard matrix of order  $2q(q + 1)$  for every prime power  $q \equiv 1 \pmod{4}$ , built from the quadratic character on  $\mathbb{F}_q$  and the  $2 \times 2$  Paley–II replacement blocks; since the Paley–II seed has order  $2(q + 1)$ , this is the natural Paley–II analogue of the lift. The orders so produced are not claimed as new existence results. Every order  $2q(q + 1)$  whose odd part is at most 3000 is already recorded in the standard tables, and the smallest order lying beyond that audited range,  $q = 81$  (order  $13284 = 4 \cdot 41 \cdot 81$ ), also follows from a classical Turyn product using  $T$ -matrices of order 41 and Williamson-type matrices of order  $81 = 9^2$ . We make the coverage explicit, exhibit the witnessing constructions, identify the smallest unaudited order, and tabulate those orders up to  $q = 2917$  for which no elementary witness is presently evident, emphasising that these are *unaudited* rather than open.

## 1 Introduction

For a prime power  $q$  let  $\mathbb{F} = \mathbb{F}_q$  and let  $\chi: \mathbb{F} \rightarrow \{0, \pm 1\}$  be the quadratic character,  $\chi(0) = 0$ ,  $\chi(x) = 1$  for nonzero squares and  $-1$  otherwise. Paley’s two constructions split on the residue of  $q$ : for  $q \equiv 3 \pmod{4}$  Paley I gives a Hadamard matrix of order  $q + 1$ , and for  $q \equiv 1 \pmod{4}$  Paley II gives one of order  $2(q + 1)$  [2].

Scarpis [3] showed that for a prime  $p \equiv 3 \pmod{4}$  a Hadamard matrix of order  $p + 1$  lifts to one of order  $p(p + 1)$ . Đoković [7] replaced the prime by an arbitrary prime power: if  $q \equiv 3 \pmod{4}$  is a prime power there is a Hadamard matrix of order  $q(q + 1)$ ; his note closes with the open problem of an analogous procedure for  $q \equiv 1 \pmod{4}$ .

Because the Paley–II seed has order  $2(q + 1)$ , the natural Scarpis-type target in the  $q \equiv 1 \pmod{4}$  case is

$$q \cdot 2(q + 1) = 2q(q + 1).$$

We construct such a matrix directly from  $\chi$ . Throughout,  $q \equiv 1 \pmod{4}$  is a prime power, so  $\chi(-1) = 1$ ; this is the only place the residue class is used, and it is exactly what fails for  $q \equiv 3 \pmod{4}$ .

## 2 The Paley–II blocks

Put

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and let  $\psi: \{0, \pm 1\} \rightarrow \{\pm 1\}^{2 \times 2}$  be  $\psi(0) = Z$ ,  $\psi(1) = P$ ,  $\psi(-1) = -P$ ; these are the Paley–II replacement blocks, and  $ZR = -P$ . Write  $\mathcal{R} = J_q \otimes I_2$ , where  $J_q$  is the  $q \times q$  all-ones matrix. We use repeatedly

$$\sum_{u \in \mathbb{F}} \psi(\chi(u)) = Z, \quad (2.1)$$

which holds because the nonzero squares and non-squares are equinumerous.

## 3 The construction

For  $t \in \mathbb{F}$  let  $K_t$  be the  $2q \times 2q$  matrix with rows and columns indexed by pairs  $(x, a)$ ,  $x \in \mathbb{F}$ ,  $a \in \{0, 1\}$ , whose  $2 \times 2$  block in position  $(x, y)$  is

$$K_t[x, y] = \psi(\chi(y - x - t)).$$

**Lemma 1.** *For every  $t \in \mathbb{F}$ ,  $K_t K_t^T = 2(q+1)I_{2q} - 2\mathcal{R}$ .*

*Proof.* The  $(x, x')$  block of  $K_t K_t^T$ , with  $d = x' - x$  and  $z = y - x - t$ , equals  $\sum_z \psi(\chi(z))\psi(\chi(z-d))^T$ . For  $d = 0$  each summand is  $2I_2$ , giving  $2qI_2$ . For  $d \neq 0$  the terms  $z = 0$  and  $z = d$  carry opposite skew parts and cancel, using  $\chi(-d) = \chi(d)$  (here  $\chi(-1) = 1$ ); for  $z \neq 0, d$ ,  $\psi(\chi(z))\psi(\chi(z-d))^T = 2\chi(z)\chi(z-d)I_2$ , and  $\sum_z \chi(z(z-d)) = -1$  yields  $-2I_2$ . These are the diagonal and off-diagonal blocks of  $2(q+1)I - 2\mathcal{R}$ .  $\square$

**Lemma 2.** *For  $r, s \in \mathbb{F}$ ,*

$$\sum_{a \in \mathbb{F}} K_{ar} K_{as}^T = \begin{cases} 2q(q+1)I_{2q} - 2q\mathcal{R}, & r = s, \\ 2\mathcal{R}, & r \neq s. \end{cases}$$

*Proof.* The case  $r = s$  is  $q$  copies of Lemma 1. For  $r \neq s$ , in the  $(x, x')$  block put  $z = y - x - ar$ ; the second argument becomes  $z - (x' - x) + a(r - s)$ , which ranges over all of  $\mathbb{F}$  independently of  $z$  as  $a$  does. By (2.1) each block is  $ZZ^T = 2I_2$ , i.e. the corresponding block of  $2\mathcal{R}$ .  $\square$

For  $r \in \mathbb{F}$  let  $B_r$  be the  $2q \times 2q$  matrix whose row  $(x, a)$  equals row  $(r, a)$  of  $K_0$  for every  $x$ ; thus  $B_r$  has only two distinct rows. From Lemma 1,  $B_r B_s^T = 2q\mathcal{R}$  if  $r = s$  and  $-2\mathcal{R}$  if  $r \neq s$ .

Form the  $2q \times 2q(q+1)$  block rows  $M_r = [B_r \mid K_{ar} \ (a \in \mathbb{F})]$ .

**Lemma 3.**  *$M_r M_s^T = 2q(q+1)I_{2q}$  if  $r = s$ , and 0 if  $r \neq s$ .*

*Proof.*  $M_r M_s^T = B_r B_s^T + \sum_a K_{ar} K_{as}^T$ . For  $r = s$  the  $2q\mathcal{R}$  of the border cancels the  $-2q\mathcal{R}$  of Lemma 2; for  $r \neq s$  the  $-2\mathcal{R}$  cancels the  $2\mathcal{R}$ .  $\square$

Let  $C$  be the Paley conference matrix indexed by  $\{\infty\} \cup \mathbb{F}$ , namely  $C_{u,u} = 0$ ,  $C_{\infty,x} = C_{x,\infty} = 1$ ,  $C_{x,y} = \chi(y-x)$ , and let  $S$  be the order- $2(q+1)$  Paley–II Hadamard matrix obtained by replacing  $0, 1, -1$  by  $Z, P, -P$ . Delete the two rows of  $S$  indexed by  $\infty$ ; in each remaining row repeat the  $\infty$ -column pair  $q$  times unchanged, and replace each finite column pair by its product with  $R$ , repeated  $q$  times. This is a  $2q \times 2q(q+1)$  matrix  $M_\infty$ .

**Lemma 4.**  $M_\infty M_\infty^T = 2q(q+1)I_{2q}$  and  $M_\infty M_r^T = 0$  for all  $r \in \mathbb{F}$ .

*Proof.* The first identity holds because  $S$  is Hadamard, each column pair is repeated  $q$  times, and right multiplication by  $R$  preserves row inner products. For the cross identity, summing a cap row against a row of  $M_r$  and using (2.1) on every  $2q$ -block leaves  $(P_\alpha + (\sum_a S[e, a]_\alpha)R) \cdot Z_\beta$ ; the inner sum of finite entries of a row of  $S$  is  $Z_\alpha$ , and  $P + ZR = P - P = 0$ .  $\square$

**Theorem 1.** *Let  $q \equiv 1 \pmod{4}$  be a prime power. Then*

$$H = \begin{bmatrix} M_\infty \\ M_r \ (r \in \mathbb{F}) \end{bmatrix}$$

*is a Hadamard matrix of order  $2q(q+1)$ .*

*Proof.*  $H$  is  $\{1, -1\}$  and square of order  $2q + q \cdot 2q = 2q(q+1)$ . Lemmas 3 and 4 give  $HH^T = 2q(q+1)I$ .  $\square$

**Corollary 1.** *For every prime power  $q \equiv 1 \pmod{4}$  there is a Hadamard matrix of order  $2q(q+1)$ ; the smallest non-prime instance is  $q = 9$ , of order 180.*

The construction has been verified by full Gram computation for all  $q \leq 49$  (orders 60,  $\dots$ , 4900, including the extension fields  $q = 9, 25, 49$ ) and by streamed row-orthogonality checks for  $q = 81$  and  $q = 125$  (orders 13284 and 31500).

## 4 Remarks on novelty

Theorem 1 supplies, for the  $q \equiv 1 \pmod{4}$  case, the uniform family that Đoković [7] asked for. It does not appear to resolve previously open Hadamard orders; its contribution is the uniform construction, and we record the evidence for that framing.

Since  $q \equiv 1 \pmod{4}$ , the 2-adic valuation of the order is exactly 2:

$$2q(q+1) = 4 \cdot q \cdot \frac{q+1}{2}, \quad q \text{ and } \frac{q+1}{2} \text{ both odd.} \quad (4.1)$$

Thus every order is  $4 \times (\text{odd})$ , so it is never a Kronecker product of two smaller Hadamard matrices, and its odd part  $q(q+1)/2$  is always composite. Two consequences follow. First, the family never meets the dominant list of open Hadamard orders, which are  $4 \times \text{prime}$ . Second, (4.1) has the form needed for Turyn's construction: when  $T$ -matrices of order  $t$  and Williamson-type matrices of order  $w$  are available, one obtains a Hadamard matrix of order  $4tw$  [5, 6], and here  $\{t, w\} = \{q, (q+1)/2\}$ . The coverage analysis of Appendix A makes both points quantitative: all orders with odd part  $\leq 3000$  are recorded in the standard tables, the first order beyond that range is covered by such a Turyn product, and the orders for which no elementary witness is currently evident are listed but are unaudited rather than open.

## A Coverage analysis

Write  $N = 2q(q+1)$  and  $m = q(q+1)/2$  for its odd part, so  $N = 4m$ . We assess coverage of  $N$  by prior constructions on three levels: the audited tables (§A.1), the first unaudited order (§A.2), and an elementary-witness test that isolates candidate orders (§A.3).

### A.1 Orders with odd part at most 3000

The surveys of Seberry and Yamada [6, 9] and the database of Cati and Pasechnik [10] record, for every odd  $m \leq 3000$ , the least  $t$  for which a Hadamard matrix of order  $2^t m$  is known. The default is  $t = 2$  (so  $N = 4m$  is known), and the only odd  $m \leq 3000$  with  $t > 2$  are the primes 167, 179, 223, 283 (orders 668, 716, 892, 1132, still open). Since  $m = q(q+1)/2$  is composite it is never one of these; hence *every  $N$  with  $m \leq 3000$  is a known Hadamard order*. This is the range  $q \leq 73$ , tabulated below with an explicit elementary witness where one exists (Paley I/II, or a Turyn product  $4tw$ ).

| $q$ | $N = 2q(q+1)$ | odd part $m$         | witnessing construction                  |
|-----|---------------|----------------------|------------------------------------------|
| 5   | 60            | $3 \cdot 5$          | Paley II, $p = 29$                       |
| 9   | 180           | $3^2 \cdot 5$        | Paley II, $p = 89$                       |
| 13  | 364           | $7 \cdot 13$         | Paley II, $p = 181$                      |
| 17  | 612           | $3^2 \cdot 17$       | Turyn $4 \cdot 9 \cdot 17$               |
| 25  | 1300          | $5^2 \cdot 13$       | Turyn $4 \cdot 13 \cdot 25$              |
| 29  | 1740          | $3 \cdot 5 \cdot 29$ | Turyn $4 \cdot 15 \cdot 29$              |
| 37  | 2812          | $19 \cdot 37$        | Turyn $4 \cdot 19 \cdot 37$              |
| 41  | 3444          | $3 \cdot 7 \cdot 41$ | Paley II, $p = 1721$                     |
| 49  | 4900          | $5^2 \cdot 7^2$      | Turyn $4 \cdot 25 \cdot 49$              |
| 53  | 5724          | $3^3 \cdot 53$       | Paley II, $p = 2861$                     |
| 61  | 7564          | $31 \cdot 61$        | Turyn $4 \cdot 31 \cdot 61$              |
| 73  | 10804         | $37 \cdot 73$        | tables; no elementary witness (see §A.3) |

The entry  $q = 73$  is instructive: its odd part  $2701 = 37 \cdot 73$  carries the same obstruction we use below (the prime 73), yet  $N = 10804$  is a known Hadamard order because  $m \leq 3000$  is audited. Obstruction in the elementary test is therefore a property of that test, not of Hadamard existence.

### A.2 The first unaudited order: $q = 81$

The smallest order of the family whose odd part exceeds 3000, and hence lies beyond the audited range, is

$$q = 81 = 3^4, \quad N = 13284, \quad m = 3321 = 3^4 \cdot 41 > 3000.$$

It is nevertheless a known Hadamard order. By (4.1),

$$13284 = 4 \cdot 41 \cdot 81 = 4tw, \quad t = 41, \quad w = 81,$$

and both ingredients are classical:  $T$ -matrices of order 41 come from the base sequences  $BS(21, 20)$  [5, 8], and Williamson-type matrices of order  $81 = 9^2$  come from Turyn's 9-power family [5, 6]. The Turyn array on these [6, 9] yields a Hadamard matrix of order  $4 \cdot 41 \cdot 81 = 13284$ . Thus  $q = 81$  is the smallest unaudited order, and it is covered.

### A.3 An elementary-witness test and the candidate orders

Beyond the audited range we have no table to appeal to, so we apply a deliberately conservative *elementary-witness* test based on (4.1) and Turyn's array. Call an odd prime  $p$  *T-eligible* if  $p \leq 59$  (so that  $T$ -matrices of order  $p$  follow from base sequences  $BS(m+1, m)$ ,  $m \leq 29$ ) and *W-eligible* if  $p \leq 33$  or  $2p - 1$  is a prime power (Williamson-type matrices of order  $p$ , the latter by Turyn's  $(P+1)/2$  family with  $P = 2p - 1 \equiv 1 \pmod{4}$ ); both classes are multiplicatively closed. An order  $N = 4m$  passes the test, and is then a Hadamard order, if  $N$  is a Paley I/II order or if every prime

factor of  $m$  is  $T$ - or  $W$ -eligible. A prime factor  $p$  of  $m$  is *obstructing* if  $p > 59$  and  $2p - 1$  is not a prime power.

A passing verdict is a genuine construction; a failing verdict (*no elementary witness*) is a statement about this test only, not a claim that  $N$  is open. Over the 232 orders with  $q \leq 3000$ , the test passes 131 (audited, Paley, or Turyn) and fails 101. The smallest failure is

$$q = 109, \quad N = 23980, \quad m = 5 \cdot 11 \cdot 109,$$

obstructed by 109 ( $2 \cdot 109 - 1 = 217 = 7 \cdot 31$  is not a prime power). This  $N = 23980$  is the smallest order of the family for which we exhibit neither an audited entry nor an elementary witness; we do not assert it is open, and indeed Williamson-type matrices of order 109, were they tabulated, would settle it at once (compare  $q = 73$ ).

Table 1 lists all such candidate orders up to  $q = 2917$ . The previously discussed  $q = 2689$  and  $q = 2917$  appear at the end; both are obstructed (2689 and 2917 are primes with  $2 \cdot 2689 - 1 = 5377 = 19 \cdot 283$  and  $2 \cdot 2917 - 1 = 5833 = 19 \cdot 307$ ), and  $q = 2917$  has the extreme form  $m = 1459 \cdot 2917$ , a product of two obstructing primes. They are not distinguished among the candidates except by size.

Table 1: Orders  $N = 2q(q + 1)$ ,  $q$  a prime power  $\equiv 1 \pmod{4}$ , with no elementary witness (§A.3),  $109 \leq q \leq 2917$ . “obstr.” lists the obstructing prime factors of the odd part  $m = q(q + 1)/2$ .

| $q$  | $N$     | odd part $m$                    | obstr.    |
|------|---------|---------------------------------|-----------|
| 109  | 23980   | $5 \cdot 11 \cdot 109$          | 109       |
| 173  | 60204   | $3 \cdot 29 \cdot 173$          | 173       |
| 197  | 78012   | $3^2 \cdot 11 \cdot 197$        | 197       |
| 277  | 154012  | $139 \cdot 277$                 | 277       |
| 389  | 303420  | $3 \cdot 5 \cdot 13 \cdot 389$  | 389       |
| 397  | 316012  | $199 \cdot 397$                 | 397       |
| 409  | 335380  | $5 \cdot 41 \cdot 409$          | 409       |
| 457  | 418612  | $229 \cdot 457$                 | 457       |
| 509  | 519180  | $3 \cdot 5 \cdot 17 \cdot 509$  | 509       |
| 521  | 543924  | $3^2 \cdot 29 \cdot 521$        | 521       |
| 593  | 704484  | $3^3 \cdot 11 \cdot 593$        | 593       |
| 613  | 752764  | $307 \cdot 613$                 | 613       |
| 617  | 762612  | $3 \cdot 103 \cdot 617$         | 103, 617  |
| 641  | 823044  | $3 \cdot 107 \cdot 641$         | 107, 641  |
| 701  | 984204  | $3^3 \cdot 13 \cdot 701$        | 701       |
| 733  | 1076044 | $367 \cdot 733$                 | 733       |
| 757  | 1147612 | $379 \cdot 757$                 | 757       |
| 797  | 1272012 | $3 \cdot 7 \cdot 19 \cdot 797$  | 797       |
| 821  | 1349724 | $3 \cdot 137 \cdot 821$         | 137, 821  |
| 829  | 1376140 | $5 \cdot 83 \cdot 829$          | 83        |
| 853  | 1456924 | $7 \cdot 61 \cdot 853$          | 853       |
| 937  | 1757812 | $7 \cdot 67 \cdot 937$          | 67        |
| 953  | 1818324 | $3^2 \cdot 53 \cdot 953$        | 953       |
| 1009 | 2038180 | $5 \cdot 101 \cdot 1009$        | 101       |
| 1021 | 2086924 | $7 \cdot 73 \cdot 1021$         | 73, 1021  |
| 1033 | 2136244 | $11 \cdot 47 \cdot 1033$        | 1033      |
| 1097 | 2409012 | $3^2 \cdot 61 \cdot 1097$       | 1097      |
| 1109 | 2461980 | $3 \cdot 5 \cdot 37 \cdot 1109$ | 1109      |
| 1117 | 2497612 | $13 \cdot 43 \cdot 1117$        | 1117      |
| 1129 | 2551540 | $5 \cdot 113 \cdot 1129$        | 113, 1129 |
| 1153 | 2661124 | $577 \cdot 1153$                | 1153      |
| 1181 | 2791884 | $3 \cdot 197 \cdot 1181$        | 197, 1181 |
| 1213 | 2945164 | $607 \cdot 1213$                | 1213      |

| $q$  | $N$      | odd part $m$                            | obstr.    |
|------|----------|-----------------------------------------|-----------|
| 1217 | 2964612  | $3 \cdot 7 \cdot 29 \cdot 1217$         | 1217      |
| 1249 | 3122500  | $5^4 \cdot 1249$                        | 1249      |
| 1289 | 3325620  | $3 \cdot 5 \cdot 43 \cdot 1289$         | 1289      |
| 1321 | 3492724  | $661 \cdot 1321$                        | 1321      |
| 1361 | 3707364  | $3 \cdot 227 \cdot 1361$                | 227, 1361 |
| 1373 | 3773004  | $3 \cdot 229 \cdot 1373$                | 1373      |
| 1381 | 3817084  | $691 \cdot 1381$                        | 1381      |
| 1453 | 4225324  | $727 \cdot 1453$                        | 1453      |
| 1481 | 4389684  | $3 \cdot 13 \cdot 19 \cdot 1481$        | 1481      |
| 1553 | 4826724  | $3 \cdot 7 \cdot 37 \cdot 1553$         | 1553      |
| 1597 | 5104012  | $17 \cdot 47 \cdot 1597$                | 1597      |
| 1601 | 5129604  | $3^2 \cdot 89 \cdot 1601$               | 89, 1601  |
| 1613 | 5206764  | $3 \cdot 269 \cdot 1613$                | 269, 1613 |
| 1637 | 5362812  | $3^2 \cdot 7 \cdot 13 \cdot 1637$       | 1637      |
| 1709 | 5844780  | $3^2 \cdot 5 \cdot 19 \cdot 1709$       | 1709      |
| 1721 | 5927124  | $3 \cdot 7 \cdot 41 \cdot 1721$         | 1721      |
| 1733 | 6010044  | $3 \cdot 17^2 \cdot 1733$               | 1733      |
| 1741 | 6065644  | $13 \cdot 67 \cdot 1741$                | 67        |
| 1753 | 6149524  | $877 \cdot 1753$                        | 1753      |
| 1777 | 6319012  | $7 \cdot 127 \cdot 1777$                | 127, 1777 |
| 1789 | 6404620  | $5 \cdot 179 \cdot 1789$                | 179, 1789 |
| 1873 | 7020004  | $937 \cdot 1873$                        | 1873      |
| 1877 | 7050012  | $3 \cdot 313 \cdot 1877$                | 1877      |
| 1889 | 7140420  | $3^3 \cdot 5 \cdot 7 \cdot 1889$        | 1889      |
| 1933 | 7476844  | $967 \cdot 1933$                        | 1933      |
| 1997 | 7980012  | $3^3 \cdot 37 \cdot 1997$               | 1997      |
| 2017 | 8140612  | $1009 \cdot 2017$                       | 2017      |
| 2053 | 8433724  | $13 \cdot 79 \cdot 2053$                | 2053      |
| 2081 | 8665284  | $3 \cdot 347 \cdot 2081$                | 347, 2081 |
| 2113 | 8933764  | $7 \cdot 151 \cdot 2113$                | 151, 2113 |
| 2129 | 9069540  | $3 \cdot 5 \cdot 71 \cdot 2129$         | 71, 2129  |
| 2161 | 9344164  | $23 \cdot 47 \cdot 2161$                | 2161      |
| 2213 | 9799164  | $3^3 \cdot 41 \cdot 2213$               | 2213      |
| 2269 | 10301260 | $5 \cdot 227 \cdot 2269$                | 227, 2269 |
| 2273 | 10337604 | $3 \cdot 379 \cdot 2273$                | 2273      |
| 2281 | 10410484 | $7 \cdot 163 \cdot 2281$                | 163       |
| 2293 | 10520284 | $31 \cdot 37 \cdot 2293$                | 2293      |
| 2297 | 10557012 | $3 \cdot 383 \cdot 2297$                | 383, 2297 |
| 2309 | 10667580 | $3 \cdot 5 \cdot 7 \cdot 11 \cdot 2309$ | 2309      |
| 2341 | 10965244 | $1171 \cdot 2341$                       | 2341      |
| 2357 | 11115612 | $3^2 \cdot 131 \cdot 2357$              | 131, 2357 |
| 2377 | 11305012 | $29 \cdot 41 \cdot 2377$                | 2377      |
| 2381 | 11343084 | $3 \cdot 397 \cdot 2381$                | 397, 2381 |
| 2389 | 11419420 | $5 \cdot 239 \cdot 2389$                | 239, 2389 |
| 2417 | 11688612 | $3 \cdot 13 \cdot 31 \cdot 2417$        | 2417      |
| 2437 | 11882812 | $23 \cdot 53 \cdot 2437$                | 2437      |
| 2441 | 11921844 | $3 \cdot 11 \cdot 37 \cdot 2441$        | 2441      |
| 2473 | 12236404 | $1237 \cdot 2473$                       | 2473      |
| 2477 | 12276012 | $3 \cdot 7 \cdot 59 \cdot 2477$         | 2477      |
| 2609 | 13618980 | $3^2 \cdot 5 \cdot 29 \cdot 2609$       | 2609      |
| 2621 | 13744524 | $3 \cdot 19 \cdot 23 \cdot 2621$        | 2621      |
| 2657 | 14124612 | $3 \cdot 443 \cdot 2657$                | 443, 2657 |
| 2677 | 14338012 | $13 \cdot 103 \cdot 2677$               | 103, 2677 |
| 2689 | 14466820 | $5 \cdot 269 \cdot 2689$                | 269, 2689 |
| 2693 | 14509884 | $3 \cdot 449 \cdot 2693$                | 449, 2693 |
| 2713 | 14726164 | $23 \cdot 59 \cdot 2713$                | 2713      |
| 2749 | 15119500 | $5^3 \cdot 11 \cdot 2749$               | 2749      |
| 2777 | 15429012 | $3 \cdot 463 \cdot 2777$                | 463, 2777 |

| $q$  | $N$      | odd part $m$                      | obstr.    |
|------|----------|-----------------------------------|-----------|
| 2789 | 15562620 | $3^2 \cdot 5 \cdot 31 \cdot 2789$ | 2789      |
| 2797 | 15652012 | $1399 \cdot 2797$                 | 2797      |
| 2801 | 15696804 | $3 \cdot 467 \cdot 2801$          | 467, 2801 |
| 2833 | 16057444 | $13 \cdot 109 \cdot 2833$         | 109, 2833 |
| 2837 | 16102812 | $3 \cdot 11 \cdot 43 \cdot 2837$  | 2837      |
| 2857 | 16330612 | $1429 \cdot 2857$                 | 2857      |
| 2909 | 16930380 | $3 \cdot 5 \cdot 97 \cdot 2909$   | 2909      |
| 2917 | 17023612 | $1459 \cdot 2917$                 | 2917      |

None of the orders in Table 1 is a known open Hadamard order; by (4.1) each odd part is composite, while the open orders below  $10^4$  are  $4 \times$  prime. Widening the ingredient set (computer-tabulated Williamson and  $T$ -matrices, or Goethals–Seidel cyclotomic difference sets) removes candidates from the table but can never produce one that meets the open list. The role of Theorem 1 is thus to give a single closed-form rule for the whole family, including the region these tables do not reach, rather than to settle any open existence question.

## References

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